

Task 1.1: Cyclic Coordinate Descent (CCD)

Objective

In this assignment you will implement and test two variants of the *Cyclic Coordinate Descent* method:

- (a) The basic CCD algorithm (line search along coordinate directions).
- (b) The accelerated CCD algorithm with an additional acceleration step.

The goal is to understand how coordinate-wise search works, how acceleration affects convergence, and how the choice of a starting point influences the optimization process.

Algorithms to Implement

You must translate the pseudocode given in lectures into Python. The pseudocode for both algorithms is given below for reference.

$basis(i, n) = [k == i ? 1.0 : 0.0 \text{ for } k \text{ in } 1 : n]$ # basis vector in i -th direction

Algorithm 1 Cyclic Coordinate Descent

```
1: function CYCLIC_COORDINATE_DESCENT( $f, x, \epsilon$ )
2:    $\Delta = \infty, n = \text{length}(x)$ 
3:   while  $\Delta > \epsilon$  do
4:      $x' = \text{copy}(x)$ 
5:     for  $i = 1$  to  $n$  do
6:        $d = basis(i, n)$ 
7:        $x = \text{line\_search}(f, x, d)$ 
8:     end for
9:      $\Delta = \|x - x'\|$ 
10:  end while
11:  return  $x$ 
12: end function
```

Algorithm 2 Cyclic Coordinate Descent with Acceleration Step

```
1: function CYCLIC_COORDINATE_DESCENT_WITH_ACCELERATION( $f, x, \epsilon$ )
2:    $\Delta = \infty, n = \text{length}(x)$ 
3:   while  $\Delta > \epsilon$  do
4:      $x' = \text{copy}(x)$ 
5:     for  $i = 1$  to  $n$  do
6:        $d = basis(i, n)$ 
7:        $x = \text{line\_search}(f, x, d)$ 
8:     end for
9:      $x = \text{line\_search}(f, x, x - x')$  ▷ acceleration step
10:     $\Delta = \|x - x'\|$ 
11:  end while
12:  return  $x$ 
13: end function
```

Test Functions

You will apply both algorithms to the following benchmark functions:

- **Booth function (2D):**

$$f(x, y) = (x + 2y - 7)^2 + (2x + y - 5)^2$$

This function has a global minimum at $(1, 3)$ with $f(1, 3) = 0$.

- **Rosenbrock function (2D):**

$$f(x, y) = (1 - x)^2 + 5(y - x^2)^2$$

This is a classic non-convex test function with a global minimum at $(1, 1)$ where $f(1, 1) = 0$.

- **Pathological function (2D):**

$$f(x, y) = \sin(5y)\cos(5x) + x^2 + y^2$$

This function has global minima a global minimum at $(0, 0)$ where $f(0, 0) = 0$.

Starting Points

Run both algorithms from the following initial points:

- Booth function: $x^{(0)} = (0, 0)$ and $x^{(0)} = (-3, 4)$.
- Rosenbrock function: $x^{(0)} = (-1.5, 2.0)$ and $x^{(0)} = (2.0, -1.0)$.
- Pathological function: $x^{(0)} = (2.0, 2.0)$ and $x^{(0)} = (-2.0, -2.0)$.

Each combination of (algorithm, function, starting point) requires a single deterministic run.

Tasks

1. Implement both CCD algorithms in Python, following the pseudocode.
2. Implement a one-dimensional line search routine (you may use a simple method such as golden-section search or backtracking).
3. Run experiments on the specified test functions and starting points.
4. Record the final solution point, the function value at this point, and the number of iterations.
5. Produce contour plots of the objective functions with the trajectory of the optimization steps superimposed.
6. Summarize your results in a table and provide short comments:
 - Did the algorithm converge to the global minimum?
 - How did the acceleration step affect convergence?
 - How sensitive is the method to the choice of the starting point?
 - Explain why the methods converge well for some test functions and not for others. What affects convergence?

Deliverables

- Python code implementing both algorithms and the line search.
- Plots showing convergence trajectories on the Booth and Rosenbrock functions.
- A summary table (in the console) with results for all runs.
- A short written discussion (3–5 sentences) answering the questions above.